UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2023/2024

SEPTEMBER EXAMINATION

UCCN1213 FUNDAMENTALS OF COMPUTER AND INFORMATION SECURITY / UCCN1223 CYBERSECURITY

MONDAY, 9 OCTOBER 2023 TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the FIRST (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

SECTION A (Answer ALL questions)

Q1. (a) Encrypt the following message with the given key using Caesar Cipher.

M = I AM BETTER IN CYBERSECURITY.
K = 5 (4 marks)

(b) Decrypt the following ciphertext with the given key Caesar Cipher.

C = VHFXULWB LV JRRG
K = 3 (4 marks)

(c) Encipher and decipher the following words using a Keyword Cipher with the keyword **WIRELESS**.

(i) RESEARCH (2 marks)

(ii) PQTELJQP (2 marks)

(d) Encipher and decipher the following words using a Keyword Mixed Cipher with the keyword **PREFERENCE**.

(i) UTAR MY CHOICE (2 marks)

(ii) QYPK (2 marks)

(e) Encipher and decipher the following words using a keyword Transposed Cipher with the keyword **PUTRAJAYA**.

(i) FICT MY FACULTY (2 marks)

(ii) OLJAU (2 marks)

(f) Encipher and decipher the following words using an Interrupted Keyword Cipher with the keyword **WIRELESS**.

(i) I WILL PASS THE MARK (3 marks)

(ii) PKOFI (2 marks)
[Total : 25 marks]

Q2. (a) Distinguish between Denial-of-Service Attacks (DoS) and Distributed Denial-of-Service Attacks (DDoS) and list out any two motivations behind the attacks. (4 marks)

(b) Explain the **THREE (3)** key security concepts and their levels of impact. (7 marks)

(c) Draw and **EXPLAIN** the network security model. (6 marks)

(d) Briefly, discuss the following terms.

(i) Plaintext (1 mark)

(ii) Key (1 mark)

(iii) Encryption algorithm (1 mark)

(iv) Decryption algorithm (1 mark)

(e) Differentiate between substitution and transposition. (4 marks)

[Total : 25 marks]

Q3. (a) In asymmetric encryption, list and **EXPLAIN TWO (2)** key pairs used for encryption and decryption. (4 marks)

(b) Alice and Bob have generated a public and private key pair. They do not know each other's keys yet. They want to exchange messages in M over the network.

(i) What is the procedure to exchange the message confidentiality, if only passive attacks need to be considered? (2 marks)

(ii) What is the main threat if active attacks are possible over a network? (2 marks)

(iii) Suggest **TWO (2)** ways to ensure public keys' authenticity to avoid the risk of active attacks. (2 marks)

(c) The virus has **FOUR (4)** phases. Briefly described the phases of the virus. (4 marks)

Q3. (Continued)

(d) There are **FOUR (4)** potential types of attacks in network security. Briefly explain the types of attacks. (4 marks)

(e) A firewall is designed to prevent unauthorized electronic access to a network computer system. List out the **TWO (2)** types of firewalls. (2 marks)

(f) Ms. Ruwaida designs a new security model using an IoT application in her hospital. After a few months, she suspects attacks on the application. As a security manager, what kind of **BLOCKCHAIN** technology would you suggest to prevent these attacks? (5 marks)

[Total : 25 marks]

SECTION B (Choose only ONE question)

Q4. (a) Give **SIX (6)** reasons to conduct computer forensics. (6 marks)

(b) Describe **FOUR (4)** types of evidence in computer forensics. (8 marks)

(c) If evidence is used in court proceedings or actions that could be challenged legally, the evidence must meet certain standards. Identify these **THREE (3)** standards of evidence. (6 marks)

(d) Assume you are a Police Forensic Investigator investigating a homicide case. You receive a desktop computer as one of the pieces of evidence. You suspect some of the files in the computer had been deleted. Have the files really been deleted? Justify your answer. (5 marks)

[Total : 25 marks]

Q5. (a) Described **FOUR (4)** possible strategies to control risk. (8 marks)

Q5. (Continued)

(b) TechCo, a leading technology company specializing in software development, experienced a sophisticated cyber-attack. Hackers infiltrated the company's network through a phishing email sent to an unsuspecting employee. The attacker gained access to the company's internal systems, compromising a significant amount of sensitive intellectual property, including proprietary source code, trade secrets, and confidential client data.

(i) What are the incident response and mitigation of the above scenario? (3 marks)

(ii) Suggest **FIVE (5)** ways to enhance the above security. (5 marks)

(c) Draw the hierarchy of risk management components and explain any **TWO (2)** of the components. (6 marks)

(d) List out **THREE (3)** ways an organization handles incidents. (3 marks)

[Total : 25 marks]
